

Resources and Trade: The Heckscher-Ohlin Model

ECON 308 International Economic Relations | Chapter 5

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Fall 2026

Opening Puzzle: Why Does Canada Export Oil to China?

- Canada and China use similar production technologies. The Ricardian model predicts *little* trade between such countries.
- Yet Canada exports **oil, potash, and lumber** to China and imports **electronics and consumer goods** — billions of dollars each year.
- **The puzzle:** if technology is the same, what drives such different trade patterns?
- **Answer:** factor *endowments*. Canada has vast land and resources; China has abundant unskilled labour.

Today's Model

The **Heckscher-Ohlin model** shows how differences in factor endowments generate comparative advantage even with identical technology.

Country	Abundant factor	Main exports
Canada	Land, resources	Oil, wheat, lumber
China	Unskilled labour	Electronics, textiles
USA	Capital, skills	Aircraft, software
Germany	Skilled labour	Machinery, cars
Bangladesh	Unskilled labour	Garments

Ricardian vs. H-O:

Ricardian: technology differences

H-O: factor endowment differences

Both generate comparative advantage but through different mechanisms.

Where We Are: The Model Family

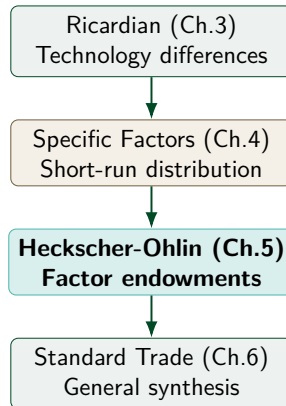
- **Ch. 2** (Lecture 1): Trade facts and the gravity model — *who* trades with whom.
- **Ch. 3** (Lecture 2): *Why* trade? Technology differences (Ricardian). Everyone gains.
- **Ch. 4** (Lecture 3): *Who* wins and loses? Short-run specificity. Some factors lose.
- **Ch. 5** (today): Long-run H-O model. Both factors mobile. *What* drives trade when technology is the same everywhere?
- **Ch. 6** (next): The Standard Trade Model — a general synthesis.

Short run vs. long run:

Ch. 4 (Specific Factors) = capital locked in sector.

Ch. 5 (H-O) = capital *fully mobile* across sectors.

As time passes, SF $\rightarrow$ H-O.



← today

Learning Objectives

After this lecture you will be able to:

1. Define **factor abundance** and **factor intensity**; state the Heckscher-Ohlin theorem.
2. Derive cost-minimizing factor ratios from Cobb-Douglas technology and determine comparative advantage from factor endowments.
3. Apply the **Stolper-Samuelson theorem** to identify winners and losers from trade in the long run.
4. Explain the **magnification effect** and show why real factor gains/losses exceed the price change.
5. State the **Factor Price Equalization** theorem and explain why it fails empirically.
6. Apply the **Rybczynski theorem** to predict how factor accumulation alters the production mix.
7. Connect the Rybczynski theorem to **Dutch Disease** and Canadian immigration policy.
8. Evaluate the **Leontief Paradox** and the trade-versus-technology debate on wage inequality.

Krugman, Obstfeld & Melitz, *International Economics*, 12th ed., Chapter 5.

The $2 \times 2 \times 2$ Framework

The H-O model assumes:

- **2 countries:** Home (H), Foreign (F).
- **2 goods:** Computers (capital-intensive), Textiles (labour-intensive).
- **2 factors:** Capital K and Labour L — *both fully mobile* across sectors (long run).
- **Same technology:** identical production functions in both countries.
- **Identical homothetic preferences.**
- **Perfect competition** everywhere.

The Single Difference

The *only* difference is factor endowments: $(K/L)_H \neq (K/L)_F$. This alone drives trade.

Variable list:

K, L = capital and labour (Home)
 K^*, L^* = capital and labour (Foreign)
 a_{KX}, a_{LX} = K and L per unit of X
 a_{KY}, a_{LY} = K and L per unit of Y
 w = wage rate
 r = rental rate of capital
 P_X, P_Y = goods prices (world)

Key ratio:

$(K/L)_H > (K/L)_F$
 $\Rightarrow$ Home is **capital-abundant**
 $\Rightarrow$ Foreign is **labour-abundant**

Factor Abundance: Relative, Not Absolute

Definition: Factor Abundance

Country H is **capital-abundant** relative to country F if:

$$\frac{K_H}{L_H} > \frac{K_F}{L_F}$$

It does *not* matter that H might have fewer total workers or less capital than F in absolute terms.

Key insight: abundance is always *relative*.

- Canada has fewer workers than China, but Canada has far more capital and natural resources per worker.
- Bangladesh has fewer workers than Germany, but Bangladesh's labour-to-capital ratio is much higher.

Country	K	L	K/L
Canada (Home)	600	300	2.0
Mexico (Foreign)	300	600	0.5

Canada has **twice** the capital per worker as Mexico, even though Mexico has **twice** the total labour.

⇒ Canada is **capital-abundant**.

⇒ Mexico is **labour-abundant**.

These are the numbers from Tutorial 1 — we will return to them in our worked example.

Factor Intensity: The Other Key Concept

Definition: Factor Intensity

Good X is **capital-intensive** relative to Y if at *any* factor price ratio w/r : $K_X/L_X > K_Y/L_Y$
(Holds for all w/r — no factor intensity reversals.)

Cobb-Douglas cost minimization:

$$\frac{K_i}{L_i} = \frac{\alpha_i}{1 - \alpha_i} \cdot \frac{w}{r}$$

Computers ($\alpha_X = 0.6$): $K/L = 1.5 (w/r)$

Textiles ($\alpha_Y = 0.3$): $K/L = 0.43 (w/r)$

Computers always use more capital per worker.

Industry	K/L ratio	Capital-int.?
Semiconductors	Very high	✓
Aircraft	High	✓
Auto assembly	Medium	✓
Forestry	Medium-low	Depends
Apparel/textiles	Low	✗
Call centres	Very low	✗

Intuition

A semiconductor fab needs expensive equipment and few workers. A garment factory needs many sewing operators and little machinery. This *persistent* difference in capital-per-worker is what makes H-O work.

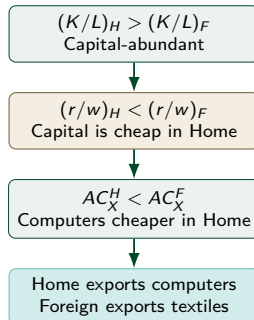
The Heckscher-Ohlin Theorem

H-O Theorem (Heckscher 1919; Ohlin 1933)

A country **exports** the good that uses its **abundant factor** intensively and **imports** the good that uses its **scarce factor** intensively.

The mechanism (step by step):

1. Home is capital-abundant: $(K/L)_H > (K/L)_F$.
2. Abundance of capital $\Rightarrow$ capital is relatively cheap in Home: $(r/w)_H < (r/w)_F$.
3. Cheap capital $\Rightarrow$ computers (capital-intensive) are *cheaper to produce* in Home.
4. Lower relative cost $\Rightarrow$ Home has **comparative advantage in computers**.
5. Home exports computers; Foreign exports textiles.



Canada: H-O Confirmed

Canada (capital/resource abundant) exports oil, potash, lumber, and wheat.

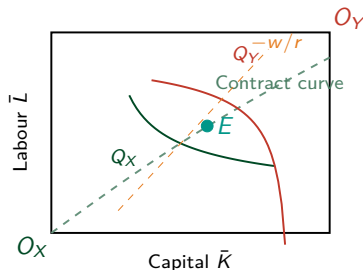
China (labour-abundant) exports garments, electronics, and assembly goods.

The pattern matches H-O predictions.

The Factor Box: How Endowments Shape Production

The factor-allocation box (Edgeworth box):

- Width = total capital $\bar{K}$; Height = total labour $\bar{L}$.
- Allocation of (K_X, L_X) to computers is read from bottom-left origin O_X .
- Remaining (K_Y, L_Y) to textiles is read from top-right origin O_Y .
- At equilibrium, both sectors are on their *isoquants* with the *same* factor price ratio (parallel tangents).



Different endowments shift the box shape. A capital-abundant country's box is *wider* relative to its height — pushing the equilibrium allocation toward more computer production.

Worked Example: Canada vs. Mexico

Worked Example

Technology (identical in both countries): $Q_C = K_C^{0.6} L_C^{0.4}$ (Computers, capital-intensive), $Q_T = K_T^{0.3} L_T^{0.7}$ (Textiles, labour-intensive).

Factor endowments:

Country	K	L	K/L	Abundant
Canada (Home)	600	300	2.0	Capital
Mexico (Foreign)	300	600	0.5	Labour

Step 1: Confirm factor intensity.

Cobb-Douglas cost minimization:

$$\frac{K_i}{L_i} = \frac{\alpha_i}{1 - \alpha_i} \cdot \frac{w}{r}$$

At any w/r , computers use more capital:

$$\frac{K_C}{L_C} = 1.5 \frac{w}{r} > \frac{K_T}{L_T} = 0.43 \frac{w}{r}$$

$\Rightarrow$ C is capital-intensive. ✓

Worked Example: Factor Prices and Trade Pattern

Step 2: Factor prices under autarky.

Canada has abundant capital $\Rightarrow$ capital is relatively cheap: $(w/r)_{Canada} = 1.5$.

Mexico has abundant labour $\Rightarrow$ labour is relatively cheap: $(w/r)_{Mexico} = 3.0$.

Step 3: K/L ratios in each country.

	$(K/L)_C$	$(K/L)_T$
Canada ($w/r = 1.5$)	$1.5 \times 1.5 = 2.25$	$0.43 \times 1.5 = 0.65$
Mexico ($w/r = 3.0$)	$1.5 \times 3.0 = 4.5$	$0.43 \times 3.0 = 1.29$

In both countries, computers remain more capital-intensive.

Step 4: Autarky relative prices.

Cheap capital in Canada $\Rightarrow$ computers relatively cheap in Canada:

$$(P_C/P_T)_{Canada} < (P_C/P_T)_{Mexico}$$

Step 5: H-O prediction.

$\Rightarrow$ Canada exports **computers**

$\Rightarrow$ Mexico exports **textiles**

If world price $(P_C/P_T)^W = 1.8$:

- Canada's ToT improves from 1.2 to 1.8.
- Mexico's ToT deteriorates from 2.2 to 1.8.
- Both still gain overall (Ch. 3 logic).

Stolper-Samuelson Theorem: Trade and Factor Returns

Stolper-Samuelson Theorem (1941)

An increase in the *relative price* of a good raises the **real return** to the factor used intensively in producing it and **lowers** the real return to the other factor — measured in terms of *both* goods.

Application to trade in Home (capital-abundant):

- Opening trade raises P_X/P_Y in Home (computers become more valuable abroad).
- $\uparrow P_X \Rightarrow$ real return to capital **rises** (capital is intensive in computers).
- $\uparrow P_X \Rightarrow$ real wage **falls** (labour is intensive in textiles).

Long-Run vs. Short-Run

Ch. 4 (Short run): labour's real wage effect is *ambiguous* — it depends on which good the worker consumes.

Ch. 5 (Long run): the real wage falls *unambiguously* in terms of *both* goods. This is harsher!

Why harsher in the long run? Because capital can move across sectors, intensifying the competition for labour.

Remember: This is real income, not just nominal. Even if the nominal wage rises slightly, it rises by *less* than prices — workers are worse off in real terms.

The Magnification Effect: Why Real Changes Exceed Price Changes

Intuition: when P_X rises, both sectors compete harder for capital (the factor intensive in X). Capital becomes scarce *economy-wide* and its return rises *more than proportionally*. Labour becomes relatively abundant and its return falls in *absolute* terms.

Zero-profit conditions pin wages and rentals:

$$P_X = a_{LX} w + a_{KX} r$$

$$P_Y = a_{LY} w + a_{KY} r$$

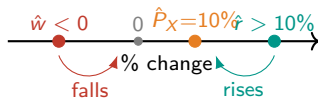
A 10% rise in P_X requires r to rise by *more than* 10% (since $a_{KX} > 0$ and r must absorb the full price increase after wages adjust).

Magnification chain (in % changes):

$$\hat{r} > \hat{P}_X > 0 > \hat{P}_Y > \hat{w}$$

Example: P_X rises 10%, P_Y unchanged.

- $\hat{r} > 10\%$: capital gains more than the price rise.
- $\hat{w} < 0$: labour loses in *absolute* terms (not just relative to computers).



Stolper-Samuelson: Worked Numerical Example

Worked Example

Given: Computers (X): $a_{LX} = 3$, $a_{KX} = 6$. Textiles (Y): $a_{LY} = 4$, $a_{KY} = 2$. Prices: $P_X = 10$, $P_Y = 5$. Find w, r ; then P_X rises to 11 (+10%).

Step 1: Initial equilibrium.

$$\text{Zero-profit: } \begin{cases} 3w + 6r = 10 \\ 4w + 2r = 5 \end{cases}$$

$$\text{Solve: } \boxed{w \approx 0.556, \quad r \approx 0.722}$$

Step 2: After $P_X = 11$ (+10%).

$$\begin{cases} 3w' + 6r' = 11 \\ 4w' + 2r' = 5 \end{cases} \Rightarrow w' \approx 0.333, \quad r' \approx 1.25$$

Step 3: Real changes.

	Before	After
P_X	10	11 (+10%)
P_Y	5	5 (0%)
w	0.556	0.333 (-40%)
r	0.722	1.25 (+73%)

Magnification Confirmed

$$\hat{r} = +73\% \gg \hat{P}_X = +10\%$$

$$\hat{w} = -40\% \ll \hat{P}_Y = 0\%$$

Active Learning: Stolper-Samuelson and Canadian Trade Policy

Discussion

Think–Pair–Share (3 minutes):

Canada is capital-abundant and opens trade with labour-abundant countries (China, Bangladesh).

1. According to Stolper-Samuelson, which Canadian factor gains from this trade? Which loses?
2. If trade lowers real wages in Canada, why do most Canadian economists *still* support free trade?
3. How does the *Specific Factors* model (Ch. 4) give a different short-run prediction about Canadian wages?

Canada–China Trade and Wages

After China joined WTO (2001), Canadian imports from China surged.

SS prediction: Canadian workers in labour-intensive sectors (textiles, apparel) should see lower real wages.

What happened? Canadian manufacturing employment fell, but wages held up — partly because displaced workers moved to services and government safety nets absorbed shocks.

Key lesson: SS is a long-run model. Short-run adjustment is painful even when long-run gains are positive.

Factor Price Equalization: Trade as a Substitute for Migration

Factor Price Equalization (Samuelson, 1948)

Under free trade with *identical technologies* and *no factor intensity reversals*, free trade in goods **equalizes factor prices internationally** — even without any factor mobility.

$$w^H = w^F, \quad r^H = r^F$$

Mechanism:

1. Free trade equalizes goods prices: $P_X^H = P_X^F$, $P_Y^H = P_Y^F$.
2. Zero-profit conditions link goods prices to factor prices: same technology + same goods prices $\Rightarrow$ same w and r .
3. Trade in goods *substitutes* for migration of factors.

Strong Prediction; Fails in Practice

FPE says US wages should equal Chinese wages under free trade. Clearly false!

Why FPE fails:

- Technology *differs* across countries.
- Trade barriers: tariffs and transport costs prevent full price equalization.
- Non-traded goods: services, housing.
- Factor intensity reversals.
- Countries may not produce the same goods (incomplete specialization).

FPE Intuition: Why Worry About Wage Equalization?

The logic of FPE:

When Canada trades with Mexico, Mexican workers do not need to move to Canada for their wages to converge. Instead:

- Canada imports textiles (labour-intensive) $\Rightarrow$ effectively “importing” Mexican labour indirectly.
- This reduces demand for Canadian unskilled labour $\Rightarrow$ Canadian wages fall toward Mexican wages.
- Mexico exports textiles $\Rightarrow$ demand for Mexican labour rises $\Rightarrow$ Mexican wages rise toward Canadian wages.

In practice: wages converge only partially and slowly due to technology differences, safety nets, and labour market institutions.

CUSMA and Wage Pressure

After NAFTA (1994), Canadian low-skilled wages did not fully equalize with Mexican wages.

Why not?

- Canada has much higher productivity (technology gap offsets FPE).
- Services sector insulates workers from goods competition.
- Canada's generous minimum wages and transfers create a wage floor.

Bottom line: FPE is a useful benchmark. Real-world convergence is partial and conditional on technology convergence.

Rybczynski Theorem: Growth and Production Bias

Rybczynski Theorem (1955)

At **constant goods prices**, an increase in one factor endowment causes the output of the good using that factor intensively to **expand** and the output of the other good to **contract**.

Mechanism:

1. Capital stock grows ($K \uparrow$) at constant prices and therefore constant w/r .
2. Each sector uses the same K/L ratio (no factor intensity reversal; w/r unchanged).
3. But more capital must be absorbed. The only way: computers (capital-intensive) expand and *release* labour that would otherwise go to textiles.
4. Textiles *contracts* because its labour supply shrinks.

Summary table:

Endowment	Q_X (K-int.)	Q_Y (L-int.)
$K \uparrow$	$\uparrow\uparrow$	$\downarrow$
$L \uparrow$	$\downarrow$	$\uparrow\uparrow$

$\uparrow\uparrow$ = super-proportional expansion

Key Implication

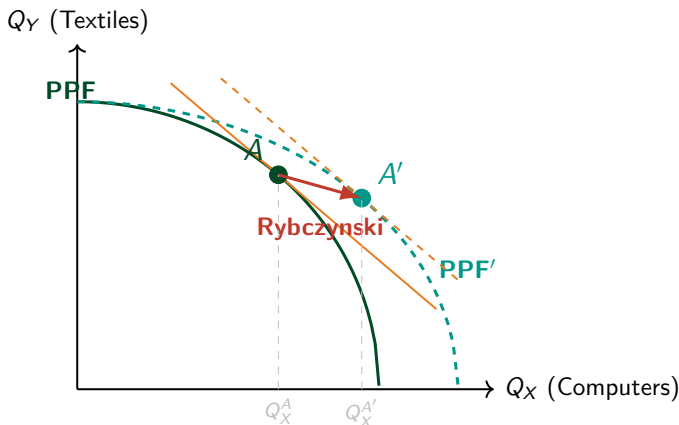
Growth reinforces comparative advantage. A capital-accumulating country becomes *even more* specialised in capital-intensive goods. Trade patterns persist and deepen.

Rybczynski: Capital Accumulation Biases the PPF

Reading the diagram:

- **PPF** (solid): before capital accumulation.
- **PPF'** (dashed): after K rises.
- Same price ratio (parallel orange lines) $\Rightarrow$ production shifts $A \rightarrow A'$.
- Q_X (computers) **expands**; Q_Y (textiles) **contracts**.
- Income rises (outer line), but gain concentrates in computers.

The **Rybczynski arrow** $A \rightarrow A'$ has a *negative slope*: one good rises, the other falls simultaneously.



Dutch Disease: Rybczynski Applied to Canada

Dutch Disease occurs when a resource boom draws factors out of manufacturing, causing manufacturing to *contract*.

1. **Resource discovery / price boom:** oil prices rise, making Alberta oil sands investment highly profitable.
2. **Capital flows in:** K in the resource sector rises.
3. **Rybczynski:** resource output expands; manufacturing output contracts (at constant goods prices).
4. **Exchange rate:** higher export revenues appreciate the CAD, making manufactured exports *even less* competitive.
5. **Double squeeze** on Ontario: higher factor costs + stronger CAD.

Alberta Oil Sands, 2002–2014

Oil investment surge: oil sands capital spending rose from \$3B/yr to \$30B/yr.

Manufacturing response: Ontario manufacturing employment fell from 1.0M (2002) to 0.72M (2014) — a 28% drop.

Exchange rate: CAD rose to parity with USD by 2007, then above parity 2009–2013.

Named after: Netherlands, whose North Sea gas discovery (1959) devastated Dutch manufacturing by 1970s.

Key insight: structural policy cannot easily offset Rybczynski-driven resource booms.

Rybczynski and Immigration: Canada's Labour Supply

Canada's immigration rate is among the highest in the OECD — over 400,000 permanent residents per year since 2022.

Rybczynski predicts:

- Immigration raises the labour supply: $L \uparrow$.
- At constant goods prices, output of labour-intensive goods expands; output of capital-intensive goods contracts.
- Canada's labour-intensive sector (services, agriculture) expands relative to capital-intensive manufacturing.

The Immigration–FPE Connection

FPE says trade substitutes for migration. If trade already partially equalized wages, immigration should have *smaller* wage effects than a closed-economy model would predict.

Discussion

Discussion question:

The Rybczynski theorem says that high immigration ($L \uparrow$) should increase output of labour-intensive goods but *reduce* output of capital-intensive goods.

1. Is this a good argument *against* immigration?
2. Does the theorem say anything about the *wage effect* on existing workers? (Hint: see SS theorem.)
3. Why might capital accumulation *accompany* immigration, moderating the Rybczynski effect?

The Leontief Paradox: Testing H-O

Leontief (1953) tested H-O with 1947 US data.

H-O prediction: US is capital-abundant $\Rightarrow$ exports should be capital-intensive.

Paradoxical finding:

- US exports were **labour-intensive**.
- US imports were **30% more capital-intensive**.
- US appeared to be a *labour-abundant* exporter — opposite of H-O!

The paradox shook confidence in H-O for decades.

Proposed resolutions:

1. **Human capital:** US exports embody skilled labour; physical capital/labour ratios miss this.
2. **Technology gap:** US comparative advantage is in technology, not physical capital per se.
3. **Tariff structure:** US tariffs protected labour-intensive industries.
4. **Demand bias:** US consumers prefer capital-intensive goods more than foreigners do.

Modern Verdict

Trefler (1995): when human capital and productivity differences are accounted for, H-O performs much better. The “missing trade” puzzle largely resolves.

Modern Evidence: Beyond Physical Capital

Key insight from empirical research: The H-O model works better when we broaden the definition of factors.

Three-factor H-O model:

- **Physical capital K :** machinery, buildings, equipment.
- **Human capital H :** education, training, skills.
- **Natural resources N :** land, oil, minerals, forests.

With all three factors:

- US: abundant in K and $H \Rightarrow$ exports aircraft, software, pharmaceuticals.
- Canada: abundant in N (and K) $\Rightarrow$ exports resources plus financial services.
- China: abundant in L (unskilled) $\Rightarrow$ exports manufactures.

	Physical K	Human H	Natural N
Canada	Medium	High	Very high
USA	High	High	Medium
China	Medium	Low-med.	Low
Germany	High	High	Low
Bangladesh	Low	Low	Low
Main export	Machinery	Soft./pharma	Resources

Canada's Multi-Factor Abundance

Canada's comparative advantage in natural resources is overwhelming. For every unit of natural capital per worker, Canada ranks among the top 3 globally. This explains the persistence of resource exports despite a high wage level.

Trade and Wage Inequality: Two Explanations

Fact: In rich countries, wage inequality has risen sharply since the 1980s.

Explanation 1 — Trade (Stolper-Samuelson):

- Trade with labour-abundant countries lowers demand for unskilled labour in capital-abundant countries.
- H-O predicts falling real wages for unskilled workers in the US and Canada.

Explanation 2 — Skill-Biased Tech Change (SBTC):

- Computers and automation displace routine/unskilled tasks.
- Raises relative demand for skilled workers.
- Works even in a closed economy.

Verdict: Both contribute, but SBTC has historically accounted for more of the rise.

Factor	Trade	SBTC
Unskilled labour	Lose	Lose
Skilled labour	Gain	Gain
Capital owners	Gain	Gain
Mechanism	SS theorem	Automation
Scope	Open economy	Any economy

Consensus (Autor et al.)

Trade and SBTC are complements, not rivals. SBTC raised skill premia everywhere. The “China shock” (post-2001) shows trade effects are larger than previously estimated, especially for local labour markets.

The China Shock: Trade and Regional Unemployment

Autor, Dorn & Hanson (2013): regional impact of Chinese import competition after WTO accession (2001).

Key findings:

- US zones exposed to Chinese competition saw lasting *employment and wage declines*.
- Workers did *not* easily relocate (H-O assumes mobile labour — violated).
- Aggregate gains positive, but distribution highly unequal and geographically concentrated.

Policy: free trade needs redistribution mechanisms to be politically sustainable.

Canada's China Shock

Canada's resource exports to China *grew* (H-O: Canadian comparative advantage in resources).

Textile/apparel sector contracted sharply.

Strong safety net (EI) cushioned adjustment.

CUSMA preserved auto/aerospace via rules of origin.

Macro Link

Trade shocks affect regional unemployment $\Rightarrow$ BoC rate decisions. The 2015 commodity/trade shock drove rates near zero through 2015–2017.

Canadian Wage Inequality: Trade and Technology

Canada's labour market, 1980–2025:

- Skill premium (university vs. high-school wages) rose by roughly 20–25 percentage points.
- Top-1% income share rose from 8% (1980) to 14% (2015), then stabilized.
- Manufacturing employment share fell from 20% (1980) to 9% (2022).

SS prediction: trade with China/Bangladesh should lower wages of unskilled Canadian workers.

- Partial confirmation: textile/apparel wages fell relative to economy-wide wages.
- But: Canadian minimum wage laws and collective agreements moderated the effect.
- SBTC also important: resource boom raised returns to capital and skilled engineers.

CUSMA and Skilled Labour

Under CUSMA, Canadian capital-intensive sectors (aerospace, automotive, financial services) are largely protected through rules of origin.

Result: Canada retains comparative advantage in capital- and skill-intensive sectors within North America, while losing labour-intensive sectors to global competition.

SS in CUSMA context: CUSMA raises P_X (capital-intensive goods) for Canada $\Rightarrow$ SS predicts capital and skills gain; unskilled labour faces competition from Mexico.

Policy Application: H-O and the Design of Trade Agreements

H-O has direct implications for trade agreement design:

1. Who pushes for free trade?

- Owners of the abundant factor lobby *for* trade (SS: they gain).
- Owners of the scarce factor lobby *against* trade (SS: they lose).

2. CUSMA rules of origin:

- Canadian auto sector: requires 75% North American content, protecting capital-intensive assembly.
- Mexican textiles: preferential tariff access to Canada, leveraging labour abundance.

3. CPTPP and resource endowments:

- Canada's abundant natural resources gain tariff-free access to Japan and Vietnam.
- Consistent with H-O: trade agreement unlocks endowment-based gains.

Canadian Dairy: SS in Action

Supply management (dairy, poultry, eggs) maintains high domestic prices via tariff-rate quotas.

H-O interpretation: Canada is not land-abundant enough to compete with New Zealand or Australia in dairy on world markets.

SS interpretation: Dairy sector owners (capital- and land-specific) lobby *against* trade because import competition would drive down returns to dairy capital.

Redistribution cost: Canadian consumers pay $\approx$ \$2B/year more for dairy products (C.D. Howe Institute, 2019).

Macroeconomic Connections: H-O and the Canadian Economy

1. Terms of trade and national income:

- Trade raises P_X/P_Y for exporter (Canada in computers; Saudi Arabia in oil).
- Rise in ToT $\Rightarrow$ real national income rises (import more with same exports).
- Canadian commodity boom 2003–2014: oil prices tripled $\Rightarrow$ ToT improved $\Rightarrow$ real income up in Alberta.

2. Factor accumulation and growth:

- Rybczynski: capital accumulation biases production toward capital-intensive goods.
- Amplifies comparative advantage over time.
- Connects to Standard Trade Model (Ch. 6): export-biased growth and terms-of-trade effects.

3. Trade, wages, and monetary policy:

The chain for Canada:

Trade $\uparrow$ capital return $\xrightarrow{SS}$ higher r in resource-rich sectors $\xrightarrow{\text{investment}}$ Rybczynski: resource output $\uparrow$, manufacturing $\downarrow$ $\xrightarrow{\text{exchange rate}}$ CAD appreciates $\xrightarrow{\text{further}}$ manufactured exports $\downarrow$ $\xrightarrow{\text{Bank of Canada}}$ one interest rate must serve two diverging sectors

One-Rate Problem

Canada's single monetary policy cannot simultaneously prevent inflation in booming Alberta and stimulate depressed Ontario — a recurring tension in Canadian macro policy.

Active Learning: Applying the Four Theorems

Discussion

Apply each theorem to Canada. Work in pairs (4 minutes).

1. **H-O:** Canada's oil sands require enormous capital investment. Does H-O predict Canada should export or import oil? What does the model miss?
2. **Stolper-Samuelson:** Canada recently signed the CPTPP with Vietnam (labour-abundant). Who in Canada gains? Who loses?
3. **FPE:** Should we expect Canadian and Vietnamese wages to equalize through CPTPP trade? What prevents this?
4. **Rybczynski:** Canada's immigration plan adds 500,000 people/yr. Predict the output effect on labour-intensive sectors.

Cheat-sheet: The four H-O theorems

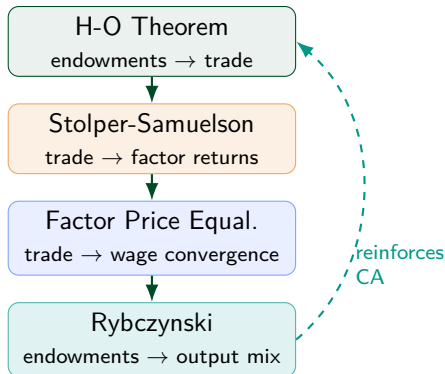
H-O: Export the good using the abundant factor intensively.

Stolper-Samuelson: Trade raises real return to the abundant factor; lowers return to the scarce factor (in terms of *both* goods).

FPE: Free trade in goods equates factor prices internationally (strong assumptions required).

Rybczynski: At constant prices, adding one factor expands output of the good intensive in that factor; contracts the other.

The Four Theorems: A Unified Picture



The four theorems work together:

- **H-O** predicts *who* trades *what*.
- **SS** predicts who *gains* and who *loses* — the political economy of trade.
- **FPE** predicts the long-run equilibrium for wages if trade is truly free.
- **Rybczynski** predicts how factor accumulation and immigration change the *production mix* and reinforce comparative advantage.

Together they form the most complete general-equilibrium model of trade we have studied.

Policy Debate: Why Do Countries Restrict Trade?

H-O makes a compelling case for free trade:

- Both countries gain in aggregate (larger consumption set).
- Abundant factors gain in real terms.
- Global resource allocation improves.

Yet governments routinely restrict trade.

H-O itself explains *why*:

- SS: scarce factor owners lose from trade.
- Scarce factors lobby for protection (concentrated, organized interests).
- Gains are diffuse; losses are concentrated.

The politics:

A few thousand steelworkers in Hamilton are a visible, organized lobby. Two million Canadian consumers each lose a few dollars from steel tariffs — they do not organize.

Steel Tariffs: SS in Practice

In 2018, the US imposed Section 232 tariffs on Canadian steel and aluminum (25% and 10% respectively).

SS winners: Canadian steelworkers (capital specific to steel protected) and steel producers.

SS losers: Canadian auto sector (input costs rose), consumers (higher prices), US consumers (welfare loss).

Retaliation: Canada imposed matching counter-tariffs on US goods.

Tariffs were lifted in May 2019. Net welfare effect: negative for both sides. Consistent with SS but also with political-economy logic.

Connecting to International Macroeconomics

H-O and the macro transmission chain for Canada:

1. **Resource endowment** (H-O): Canada exports oil, wheat, potash; China exports manufactures.
2. **Oil price boom** (terms of trade): $P_{oil} \uparrow \Rightarrow$ Canadian ToT improves $\Rightarrow$ real income rises.
3. **Rybczynski**: capital floods into Alberta oil sands; manufacturing contracts in Ontario.
4. **Exchange rate**: CAD appreciates (more demand for Canadian dollars to buy oil exports).
5. **Monetary policy**: Bank of Canada cannot cut rates (inflation in Alberta) and cannot raise rates (recession in Ontario) simultaneously.
6. **NX**: $(X - M)$ rises initially; then Dutch Disease erodes manufactured exports.

H-O in the macro framework:

$$Y = C + I + G + \underbrace{(X - M)}_{NX}$$

Trade pattern (H-O) determines the *composition* of NX: Canada's NX is positive in resources and negative in manufactured goods.

Factor price effects (SS) shape the *income distribution* — who can consume how much.

Rybczynski governs how factor accumulation shifts NX over time.

The Big Picture

H-O is not just about *what* a country trades. It determines the structure of national income, wages, and the external balance — all of which feed into macroeconomic policy.

Key Empirical Papers and Further Reading

Foundational papers:

- Heckscher, E. (1919), "The Effect of Foreign Trade on the Distribution of Income," *Ekonomisk Tidskrift*.
- Ohlin, B. (1933), *Interregional and International Trade*, Harvard Press.
- Samuelson, P. (1948), "International Trade and the Equalisation of Factor Prices," *EJ*.
- Stolper, W. & Samuelson, P. (1941), "Protection and Real Wages," *RES*.
- Rybczynski, T. (1955), "Factor Endowment and Relative Commodity Prices," *Economica*.

Empirical tests and extensions:

- Leontief, W. (1953), "Domestic Production and Foreign Trade," *PAPS*.
- Trefler, D. (1995), "The Case of the Missing Trade," *AER*, 85(5).
- Autor, D., Dorn, D., & Hanson, G. (2013), "The China Syndrome," *AER*.
- Autor, D., Dorn, D., & Hanson, G. (2016), "The China Shock," *Annual Review of Economics*.

Canadian applications:

Statistics Canada, Labour Force Survey (wage data)
Bank of Canada Working Papers on Dutch Disease
C.D. Howe Institute analyses of CUSMA and supply management

Discussion Questions

Discussion

1. Is Canada capital-abundant relative to Mexico? China? The US? How does your answer predict different trade patterns with each?
2. Rybczynski says capital accumulation hurts the labour-intensive sector. Was Alberta's oil boom bad for Ontario workers?
3. FPE predicts wages should equalize. Why don't Canadian labour unions treat this as a certainty?
4. SS says trade lowers real wages for scarce factors. Should Canada restrict imports from China? What is the efficiency trade-off?

Exam-style warm-up:

(a) Home: $K = 300$, $L = 300$. Foreign: $K = 100$, $L = 300$. Good C ($a_K = 6$, $a_L = 3$) vs. Good T ($a_K = 2$, $a_L = 4$).

1. Which country is capital-abundant?
2. Which good is capital-intensive?
3. Predict the trade pattern.
4. After trade opens, what happens to real wages in Home?

(This is Quiz 3 from Fall 2025.)

Looking Ahead: From H-O to the Standard Trade Model

What H-O gives us:

- Complete GE model: factor endowments drive trade.
- Four theorems on trade, prices, and production.
- Framework for distributional analysis and political economy.

What H-O cannot do:

- Does not explain intra-industry trade (Canada exports *and* imports cars).
- Abstracts from economies of scale.
- No welfare analysis of tariffs with terms-of-trade effects.

Chapter 6: Standard Trade Model

Unifies Ricardian, Specific Factors, and H-O as special cases.

Introduces:

- RS/RD framework for world price determination.
- Terms-of-trade welfare analysis.
- Export-biased vs. import-biased growth.
- Immiserizing growth (Bhagwati).
- Prebisch-Singer hypothesis.

Preview Question

If Canada's oil sector expands rapidly (Rybczynski), and this lowers the world price of oil (Canada's export), could Canada end up *worse off* despite growing richer? That is immiserizing growth.

Key Equations: Chapter 5 Reference Sheet

Factor abundance:

$$\frac{K_H}{L_H} > \frac{K_F}{L_F} \Rightarrow H \text{ is capital-abundant}$$

Factor intensity (Cobb-Douglas):

$$\frac{K_i}{L_i} = \frac{\alpha_i}{1-\alpha_i} \cdot \frac{w}{r}; \quad \frac{K_X}{L_X} > \frac{K_Y}{L_Y} \Rightarrow X \text{ is K-intensive}$$

H-O theorem: $(K/L)_H > (K/L)_F \Rightarrow$ Home exports X (K-intensive).

Zero-profit conditions:

$$P_X = a_{LX} w + a_{KX} r$$

$$P_Y = a_{LY} w + a_{KY} r$$

Stolper-Samuelson magnification:

$$\hat{r} > \hat{P}_X > 0 > \hat{P}_Y > \hat{w}$$

(if X is K-intensive and $P_X \uparrow$)

Factor price equalization:

Same tech + free trade $\Rightarrow w^H = w^F, r^H = r^F$

Rybczynski:

$$K \uparrow \Rightarrow Q_X \uparrow\uparrow, Q_Y \downarrow \quad (\text{at constant prices})$$

Rybczynski line:

$$\left. \frac{dQ_Y}{dQ_X} \right|_{\text{Ryb}} < 0$$

(one good rises, the other falls)

Key Takeaways

1. **H-O Theorem:** countries export goods that use their abundant factor intensively. Factor endowments drive comparative advantage when technology is identical.
2. **Stolper-Samuelson:** trade raises the real return to the abundant factor and lowers it for the scarce factor — in terms of *both* goods (magnification effect).
3. **Factor Price Equalization:** free trade in goods can equate factor prices internationally without factor migration. Requires identical technology and no barriers — too strong for the real world.
4. **Rybczynski:** at constant prices, adding a factor expands the intensive sector and *contracts* the other. Dutch Disease is Rybczynski in practice.
5. **Leontief Paradox:** H-O performs better once we expand “capital” to include human capital and technology differences.
6. **Trade and Inequality:** both trade (SS) and skill-biased technical change raise the skill premium; the China Shock shows trade effects are larger than previously estimated.
7. **Canada:** resource abundance shapes comparative advantage; Dutch Disease and SS both explain Canadian regional and wage dynamics.

Next: Chapter 6 — The Standard Trade Model

A general synthesis of all models studied so far.